Engineering Tripos Part IIB
Project Planning Form 2026/27

Students should submit this form to Group Administrators by **Friday 12 June 2026**. If plans evolve over the summer, please update and resubmit.

Name of student: Tristan Martin
Email: tjfm4@cam.ac.uk
College: Christ's
Director of Studies: Dr. Elena Punskaya

Project type: (a)

Name of supervisor: Dr. Elena Punskaya

Short title of project: Machine Learning (ML) for Atrial Fibrillation (AF) Detection and Screening

Project reference code (e.g. B-ANO22-1 / B-ANO22-type(b)): F-OP205-3

Aims of project:

This project aims to compare and evaluate state-of-the-art ML techniques for detecting AF in partnership with the SAFER study and using the public PhysioNet 2017 dataset. Each model can be evaluated using a variety of considerations ranging from accuracy to compressibility to a small device that can be used by a patient for offline inference. The project will compare modern advances in ML with those that existed in 2017 for PhysioNet's Computing in Cardiology Challenge in order to outline advances in the field and where they can be applied impactfully today.

Preparatory work for the long vacation

To prepare for this IIB project, I will begin by familiarizing myself with the mechanism of AF and current methods used for its detection/screening. Additionally, I will endeavor to understand the aims of the SAFER study to align project outcomes with the mission of the organization. From a technical standpoint, I will develop tools to visualize ECG data which will help in the work I do throughout the rest of the year. Additionally, I will research state-of-the-art ML techniques and select one or two to focus the rest of my work on.

Workshop / laboratory facilities needed

I will not require any workshop/laboratory facilities in the short term. I will get in contact with the SAFER study team and individuals at PhysioNet to bolster a relationship for future qualitative feedback.

Any special safety implications

There are little to no safety implications involved in this project as it is primarily computational. However, excessive exposure to blue light through screens and incorrect posture can lead to health issues. As such, I will ensure to work in spaced intervals and to take breaks for exercise/walks to prevent any harm.

Signature of student:

Signature of supervisor: